Week 13 – Maps with leaflet

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R script: https://github.com/Manuel-Morch/Final-Project-Bunkers/blob/c5a6fb5f0d55ca383a9b2ca6f6763b3ec0979acd/Portfolio/Week_13/StartWithLeaflet.R

DANmap final html: https://github.com/Manuel-Morch/Final-Project-Bunkers/blob/c5a6fb5f0d55ca383a9b2ca6f6763b3ec0979acd/Portfolio/Week_13/DANmap.html

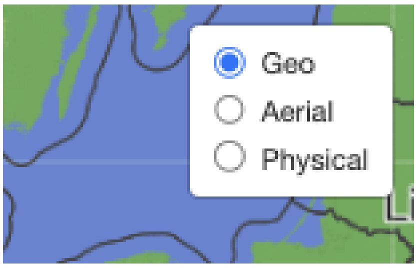
Question 1: What is the order of longitude and latitude in the setView() function?

Answer: longitude is first, then latitude.

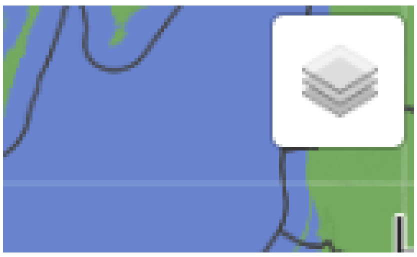
```
setView(lng = 151.005006, lat = -33.9767231, zoom = 10)
```

Question 2: How does the map above change if you replace the T in the last line of code above with F?

Answer: The layers control box is expanded by default.



Using F



Using T

Question 3: are the Latitude and Longitude columns present? Do they contain numeric decimal degrees?

Answer: Yes, they are present and contain numeric decimal degrees.

```
> glimpse(places)
Rows: 136
Columns: 8
$ Placename          <chr> "Aalborg", "Aarhus", "Aarhus", "Aarhus", "Sankt Sørens Kilde", "Møns klint", "Det Kongelige Bibliotek", "Aarhus", "Christia…
$ Type               <chr> "cultural moment", "Downtown", "Education", "Shop", "sacred place", "Natural wonder", "Library", "Shop", "Parlament", "Cult…
$ CoordinatesFILLTHIS <chr> "57.0482555,9.913457", "56.142848,10.1946182", "56.1708646,10.2040287", "56.1529896,10.2035677", "56.0800343,9.6883882", "5…
$ Latitude           <dbl> 57.04826, 56.14285, 56.17086, 56.15299, 56.08003, 54.96462, 56.17152, 56.20979, 55.67723, 55.39872, 56.13213, 55.97778, 57.…
$ Longitude          <dbl> 9.913457, 10.194618, 10.204029, 10.203568, 9.688388, 12.541842, 10.199201, 10.173500, 12.582333, 10.388611, 9.020059, 9.833…
$ Description        <chr> "Our Lady Streetart in Aalborg", "Downtown Aarhus", "Nobelparken", "Store in Aarhus", "Holy", "Cliffside", "Library", "Ikea…
$ Stars1_5           <dbl> 3, 3, 3, 5, 3, 3, 5, 1, 3, 2, 4, 3, 5, 1, 1, 4, 4, 5, 4, 2, 4, 5, 5, 2, 5, 5, 5, 3, 5, 5, 5, 1, 5, 4, 4, 3, 4, 2, 5, 3, 4, …
$ Notes              <chr> "Adela's example", NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, …
>
```

Task 1: Create a Danish equivalent of AUSmap with Esri layers but call it DANmap. You will need it layer as a background for Danish data points.

Solution:

```
l_dan <- leaflet() %>%
  setView(11.0618308,55.5357277, zoom =6)

esri <- grep("^Esri", providers, value = TRUE)

for (provider in esri) {
  l_dan <- l_dan %>% addProviderTiles(provider, group = provider)
}

DANmap <- l_dan %>%
  addLayersControl(baseGroups = names(esri),
                   options = layersControlOptions(collapsed = FALSE)) %>%
  addMiniMap(tiles = esri[[1]], toggleDisplay = TRUE,
             position = "bottomright") %>%
  addMeasure(
    position = "bottomleft",
    primaryLengthUnit = "meters",
    primaryAreaUnit = "sqmeters",
    activeColor = "#3D535D",
    completedColor = "#7D4479") %>%
  htmlwidgets::onRender("
                        function(el, x) {
                        var myMap = this;
                        myMap.on('baselayerchange',
                        function (e) {
                        myMap.minimap.changeLayer(L.tileLayer.provider(e.name));
                        })
                        }") %>%
  addControl("", position = "topright")

DANmap
```

By using the code above, I get the map below, which I can use in the following assignments

Task 2: Read in the googlesheet data you and your colleagues created into your DANmap object (with 11 background layers you created in Task 1).

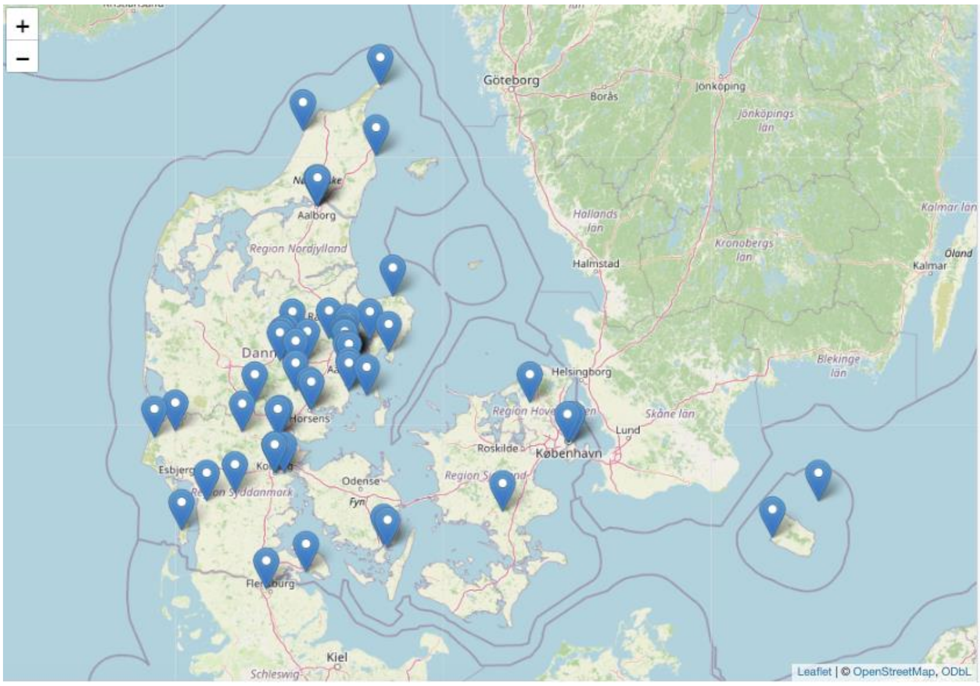
I used the following code to read in the googlesheet

```r
# Solution
places2 <- read_sheet("https://docs.google.com/spreadsheets/d/1PlxsPElZML8LZKyXbqdAYeQCDIvDps2McZx1cTVWSzI/edit?gid=1738472414#gid=1738472414",
                      col_types = "cccnncnc",
                      range = "DAM2026")

glimpse(places2)

leaflet() %>%
  addTiles() %>%
  addMarkers(lng = places2$Longitude,
             lat = places2$Latitude,
             popup = paste(places2$Placename, "<br>", places2$Description, "<br>", places2$Type))

DANmap %>%   addMarkers(lng = places2$Longitude,
                        lat = places2$Latitude,
                        popup = paste(places2$Placename,"<br>", places2$Description, "<br>", places2$Type))
```

The glimpse function shows that the information has been loaded correctly.

```
> glimpse(places2)
Rows: 76
Columns: 8
$ Placename             <chr> "Aalborg", "Bornholm", "Skævinge", "Sæby Havn", "Christania", "Thyregod Kirke", "Granola", "Bon bon land", "Agtrup Vig", "H…
$ Type                  <chr> "cultural monument", "cultural monument", "fun", "Harbour", "cultural monument", "Church", "fun", "fun", "Lookout", "Histor…
$ CoordinatesFILLTHIS   <chr> "57.0482555,9.913457", "55.1021065,14.696381", "55.9080206,12.1468987", "57.3335356,10.5305373", "55.673650209833504, 12.60…
$ Latitude              <dbl> 57.04826, 55.10211, 55.90802, 57.33354, 55.67365, 55.90779, 55.67338, 55.26093, 55.48825, 56.27933, 56.16051, 56.10531, 55.…
$ Longitude             <dbl> 9.913457, 14.696381, 12.146899, 10.530537, 12.602351, 9.258605, 12.549606, 11.863167, 9.573526, 10.037404, 10.208670, 9.684…
$ Description           <chr> "Our Lady Streetart in Aalborg", "Store Torv", "Skævinge Kro", "Harbour in Sæby with a beautiful ocean view and great ice c…
$ Stars1_5              <dbl> 3.0, 4.0, 4.3, 5.0, 4.0, 3.0, 3.9, 4.0, 4.0, NA, NA, NA, 5.0, NA, NA, NA, NA, NA, 4.3, 4.0, NA, 5.0, NA, 5.0, 2.0, 5.0, 5.0…
$ Notes                 <chr> "Adela's example", "Main Square", "Local Inn", "Beautiful ocean view", NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, …
> 
```

Below is my DANmap, with the points from the googlesheet added



Task 3: Can you cluster the points in Leaflet?

Solution

```
DANmap %>%   addMarkers(lng = places2$Longitude,
                        lat = places2$Latitude,
                        popup = paste(places2$Description, "<br>", places2$Type),
                        clusterOptions = markerClusterOptions())
```

Yes, you can cluster points using the “cluterOptions” function, as shown above

Task 4: Look at the two maps (with and without clustering) and consider what each is good for and what not.

My brief answer

The map without clustering allows you to see the exact location of each point, but it can be cluttered and hard to read if there are too many points close together. The map with clustering, groups nearby points together, making it easier to gain an overview over which areas contain more points, but it can make it harder to find individual points, especially if the map contains a lot of points.

Task 5: Find out how to display the notes and classifications column in the map.

Solution

You can display the notes and classifications column in the map by including them in the popup argument of the addMarkers() function.

```
DANmap %>%   addMarkers(lng = places2$Longitude,
                        lat = places2$Latitude,
                        popup = paste(places2$Placename,"<br>",
                                      places2$Description, "<br>",
                                      places2$Type, "<br>",
                                      places2$Notes, "<br>",
                                      places2$Classification),
                        clusterOptions = markerClusterOptions())
```